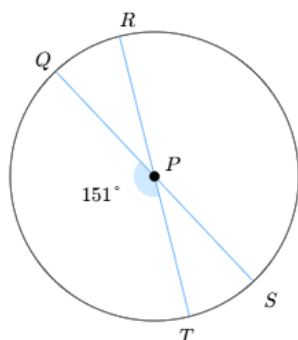


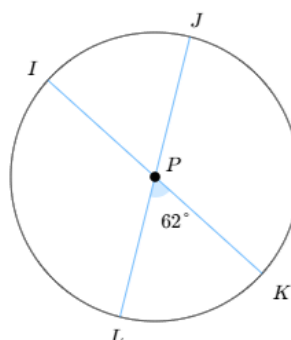
1. A circle with circumference **720** has an arc with a 5° angle. What is the length of the arc?

2. A circle with circumference **96** has an arc with a 120° angle. What is the length of the arc?

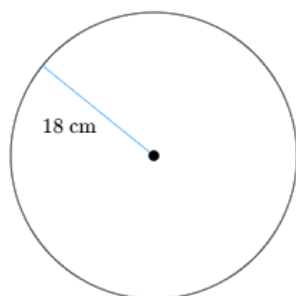
3. Find the measure of minor arc $\widehat{RS}$.



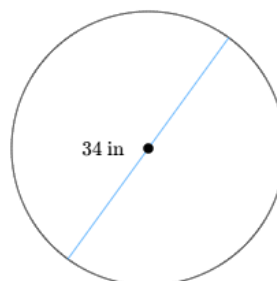
4. $\overline{IK}$ is a diameter of the circle. Find the measure of major arc $\widehat{IKL}$.



5. Find the circumference of a circle with a radius of **18** centimeters. Round your answer to the nearest tenth.

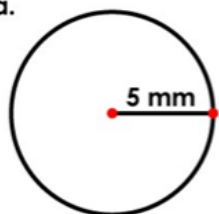


- Find the circumference of a circle with a diameter of **34** inches. Round your answer to the nearest tenth.



What is the radius and diameter of each circle?

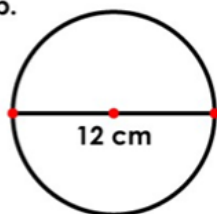
a.



radius = _____

diameter = _____

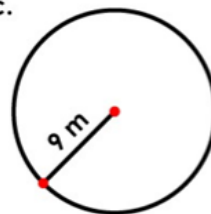
b.



radius = _____

diameter = _____

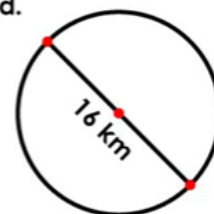
c.



radius = _____

diameter = _____

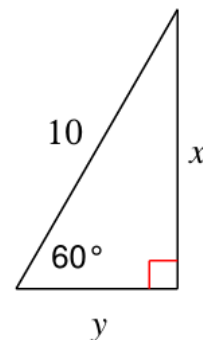
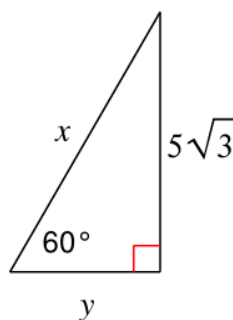
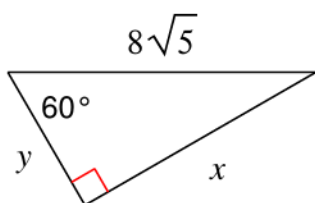
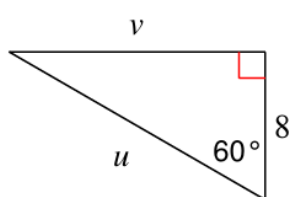
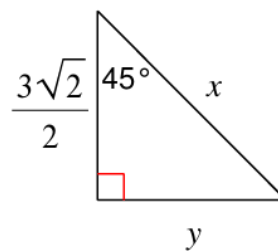
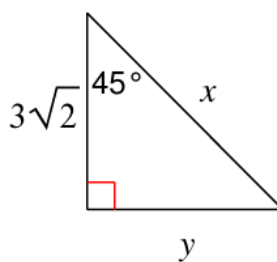
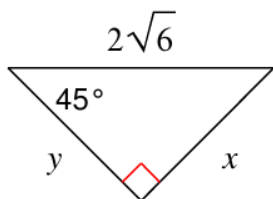
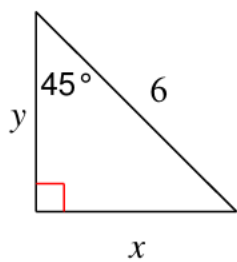
d.



radius = _____

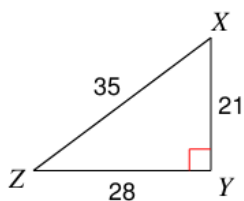
diameter = _____

- Find the missing side lengths. Leave your answers as radicals in simplest form.

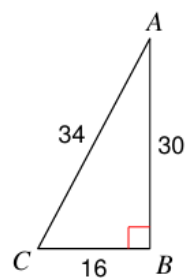


Find the value of each trigonometric ratio.

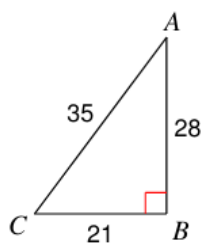
1) $\tan Z$



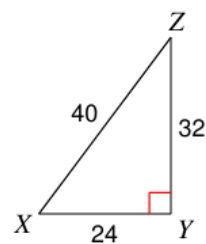
2) $\cos C$



3) $\sin C$



4) $\tan X$



- Find the missing side using Trigonometry Ratios. Round to the nearest tenth.
SOH-CAH-TOA

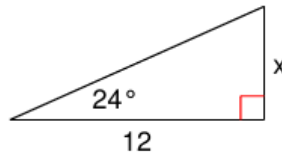
1)



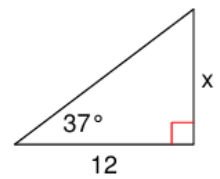
2)



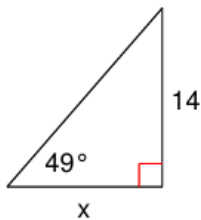
3)



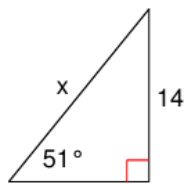
4)



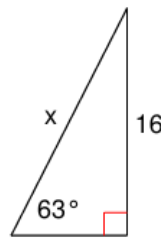
5)



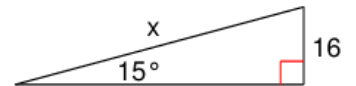
6)



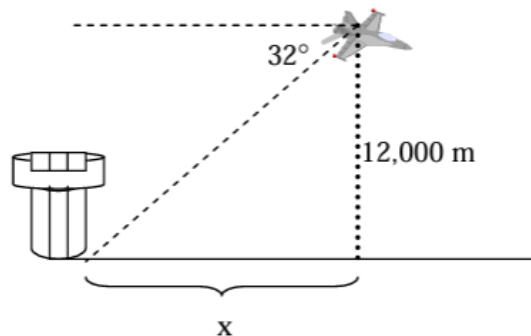
7)



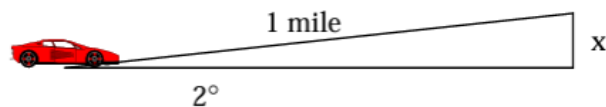
8)



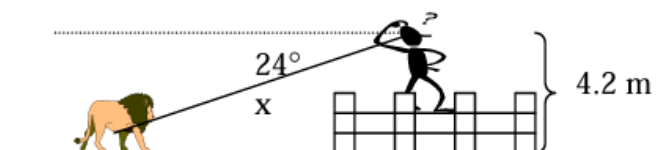
A plane is flying at an altitude of 12,000 m. From the pilot, the angle of depression to the airport tower is 32° . How far is the tower from a point directly beneath the plane?



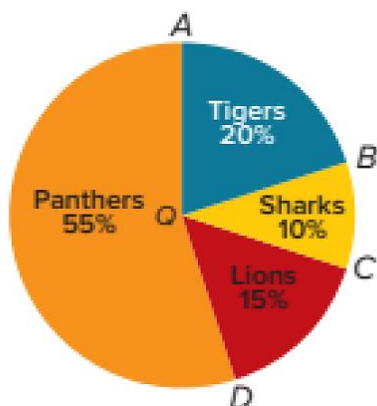
A car is traveling up a slight grade with an angle of elevation of 2° . After traveling 1 mile, what is the vertical change in feet? (1 mile = 5,280 ft)



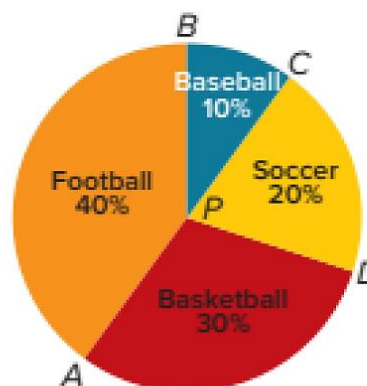
From the top of a fence, a person sites a lion on the ground at an angle of depression of 24° . If the man and the fence is 4.2 meters high, how far is the man from the lion?



SURVEYS A survey asked students at Westwood High School their preferences for the new school mascot. The results are shown in the circle graph. Find $m\widehat{AB}$.



SPORTS The circle graph shows the favorite spectator sport among a group of teens at a local high school. Find $m\widehat{AD}$.



Convert each radian measure into degrees.

• **Radians to Degrees:** Multiply the radian value by $\frac{180^\circ}{\pi}$

1) $\frac{5\pi}{12}$

2) $\frac{5\pi}{6}$

3) $\frac{\pi}{12}$

Convert each degree measure into radians.

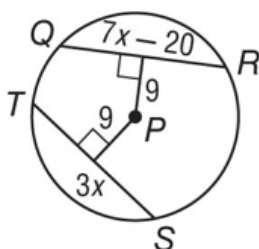
• **Degrees to Radians:** Multiply the degree value by $\frac{\pi}{180^\circ}$

1) 315°

2) 285°

3) 160°

In $\odot P$, $QR = 7x - 20$ and $TS = 3x$. What is x ?



In $\odot K$, $\overline{JL} \cong \overline{LM}$, $KN = 3x - 2$, and $KP = 2x + 1$. What is x ?



- **Note:** In trigonometry, the "ambiguous case" (SSA) occurs when you are given two sides and a non-included angle. To determine how many triangles (zero, one, or two) exist, calculate the height of the triangle: ($h = c \times \sin (A)$)

State the number of possible triangles that can be formed using the given measurements.

1) $m\angle A = 110^\circ$, $c = 19$ cm, $a = 32$ cm

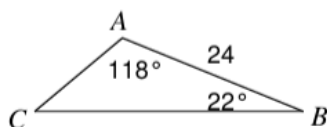
2) $m\angle A = 131^\circ$, $a = 25$ yd, $c = 8$ yd

3) $m\angle B = 100^\circ$, $a = 33$ km, $b = 29$ km

$m\angle A = 68^\circ$, $c = 34$ yd, $a = 9$ yd

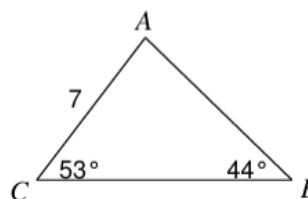
- **Find each measurement indicated using Laws of sin, Round your answers to the nearest tenth:**

1) Find AC

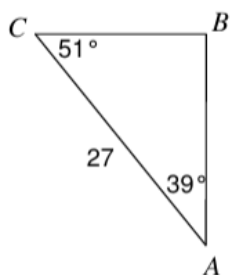


2) Find AB

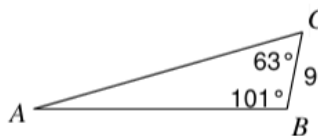
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



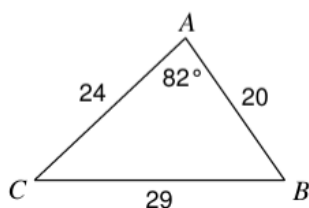
3) Find BC



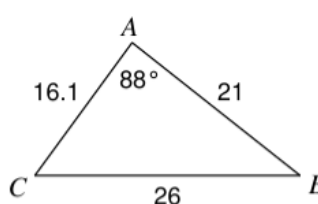
4) Find AB



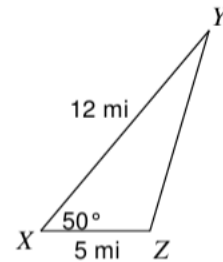
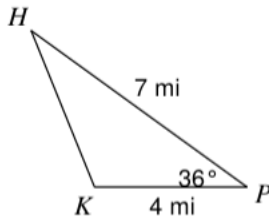
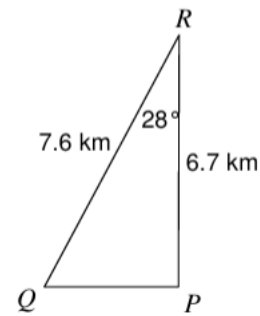
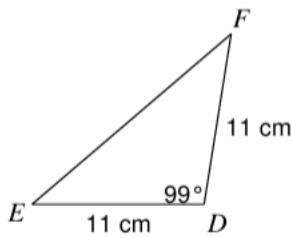
5) Find $m\angle C$



6) Find $m\angle C$



- Find the area of each triangle to the nearest tenth: $\text{Area} = \frac{1}{2} ab \sin C$



- Determine the distance and midpoint between each pair of points.

- A(1, 2, 3) and B(4, 6, 8)
- C(-2, 5, 7) and D(3, -1, 4)
- E(0, 0, 0) and F(5, 2, -3)
- G(-4, 3, 1) and H(2, -5, 6)
- I(7, -2, 4) and J(1, 3, -2)

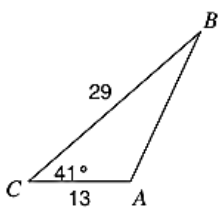
Distance Formula:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

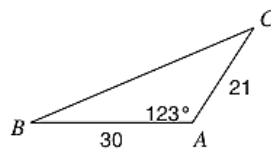
Midpoint Formula: $M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$

- Find each measurement indicated using Laws of cosines:

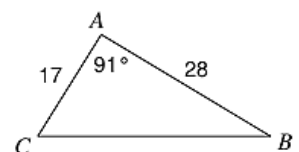
1) Find AB



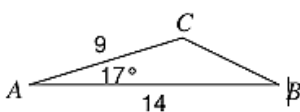
2) Find BC



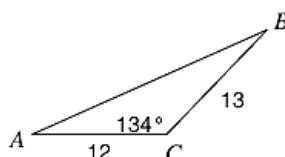
3) Find BC



4) Find BC



5) Find AB



6) Find AB

